📊 E-Commerce Sales Dashboard | Power BI

📌 Project Overview

This project is an interactive **E-Commerce Sales Dashboard** developed using **Microsoft Power BI**. The dashboard transforms raw e-commerce order data into meaningful business insights through data cleaning, transformation, analysis, and interactive visualization.

The main objective of this project is to provide a clear overview of **sales performance, orders, customers, states, and cities**, helping users understand business trends and make data-driven decisions.

🛠️ Tools & Technologies

- **Microsoft Power BI**
- **Power Query**
- **DAX**
- **Data Cleaning & Transformation**
- **Data Visualization**
- **Interactive Slicers & Filters**

🔄 Project Workflow

The complete dashboard development process followed these steps:

1️⃣ Data Collection

Collected the e-commerce sales/order dataset containing information such as:

- Order ID
- Order Date
- Customer Name
- Order State
- Order City
- Sales/Order-related information

2️⃣ Data Import

The dataset was imported into **Power BI Desktop** for analysis and visualization.

3️⃣ Data Cleaning

Using **Power Query**, the data was cleaned and prepared by:

- Removing unnecessary columns
- Handling missing values

- Correcting data types
- Removing duplicate records
- Standardizing text values
- Formatting date fields

4️⃣ Data Transformation

The cleaned data was transformed into an analysis-ready format.

Important fields were prepared for:

- Sales analysis
- Order analysis
- Customer analysis
- Location-based analysis
- Time-based analysis

5️⃣ Data Modeling

The data model was created in Power BI to establish relationships between relevant tables and enable efficient analysis.

6️⃣ DAX Calculations

DAX measures were created to calculate important business KPIs such as:

- Total Sales
- Total Orders
- Total Customers
- Average Sales
- Sales Trends

7️⃣ Dashboard Development

Interactive Power BI visuals were created to represent the data clearly.

The dashboard includes:

📌 KPI Cards
📌 Sales Analysis
📌 Order Analysis
📌 Customer Analysis
📌 State-wise Analysis
📌 City-wise Analysis
📌 Date/Time Analysis
📌 Interactive Slicers

8️⃣ Dashboard Testing

The dashboard was tested to ensure:

- Filters work correctly
- Calculations are accurate
- Visuals interact properly
- Data is represented correctly
- Dashboard navigation is user-friendly

📈 Key Dashboard Features

💰 Sales Performance

Provides an overview of overall sales performance and helps identify sales trends.

🛒 Order Analysis

Analyzes the number of orders and their distribution across different locations and time periods.

👥 Customer Analysis

Provides insights into customer activity and purchasing behavior.

📍 Geographic Analysis

Analyzes orders/sales based on:

- State
- City

📅 Time-Based Analysis

Allows users to analyze business performance across different dates and periods.

🎛️ Interactive Filters

Users can interact with the dashboard using slicers and filters to dynamically explore the data.

🖼️ Dashboard Preview

Add your Power BI dashboard screenshot inside the `images` folder and name it:

`dashboard.png`

Then use:

📂 Project Structure

```
E-Commerce-Sales-Dashboard/
│
├── 📊 E-Commerce sales Dashboard.pbix
│
├── 📁 dataset/
│   └── sales_data.csv
│
├── 📁 images/
│   └── dashboard.png
│
└── 📄 README.md
```

🎯 Project Objective

The objective of this project is to convert raw e-commerce data into an **interactive business intelligence dashboard** that provides useful insights into sales, orders, customers, and geographical performance.

The dashboard demonstrates how **Power BI, Power Query, and DAX** can be used together to support data-driven decision-making.

💡 Skills Demonstrated

- Data Analysis
- Data Cleaning
- Power Query
- DAX
- Data Modeling
- Business Intelligence
- Data Visualization
- Dashboard Development
- Interactive Reporting
- Business Insights

🚀 Future Improvements

Future versions of this project can include:

- Profit and margin analysis
- Product category analysis
- Customer segmentation
- Sales forecasting
- KPI targets
- Advanced DAX calculations
- Drill-through reports
- Power BI Service deployment

👨‍💻 Author

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